Muon Measurement Setup

Overview

The updated muon telescope (Muon Telescope V2) is based on the first version of the instrument. The key upgrade is the transition from a two-panel to a four-panel configuration, which enables track reconstruction, improved coincidence conditions, and the possibility of operating as two independent sub-telescopes.

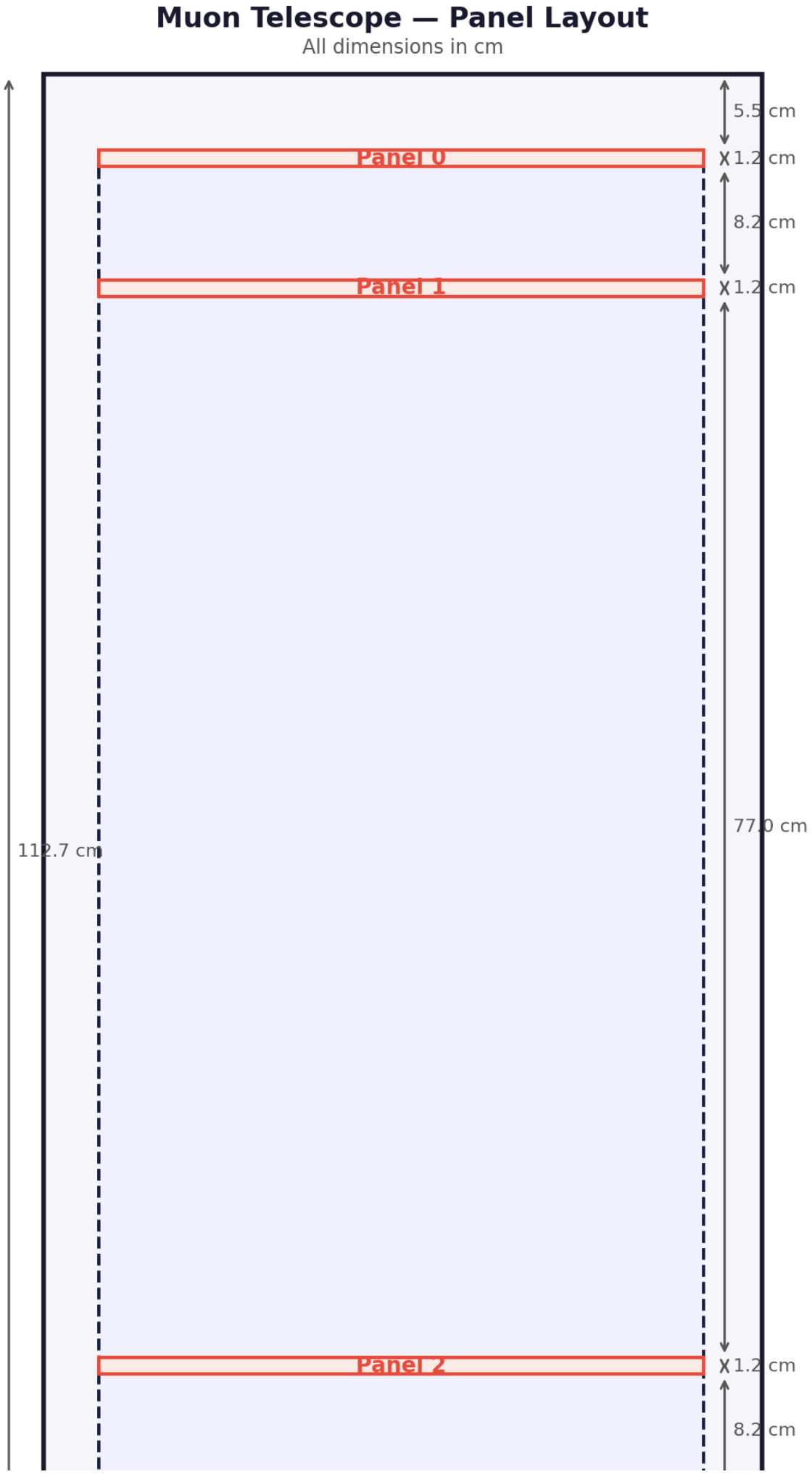
| muon_telescope_v1.png | muon_telescope_v2.png |

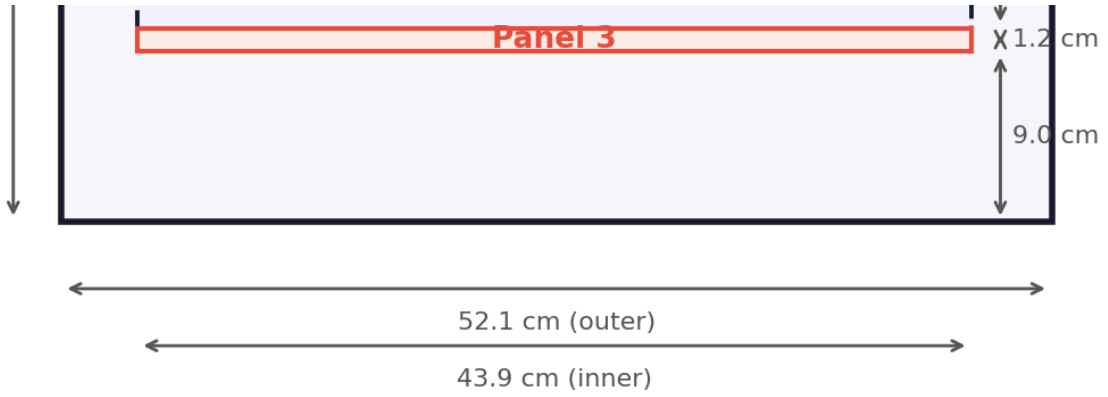
Detector Configuration

The telescope consists of four plastic scintillation panels, stacked vertically from top to bottom:

- Panel 0 (topmost)
- Panel 1
- Panel 2

- Panel 3 (bottommost)


Muon Telescope — Panel Layout
All dimensions in cm
5.5 cm
Panel 0
1.2 cm
8.2 cm
Panel 1
1.2 cm
112.7 cm
77.0 cm
Panel 2
1.2 cm
8.2 cm



Panel Specifications

- Active area: 282 mm × 282 mm = 0.0795 m² per panel
- Scintillator thickness: 10 mm
- Material: EJ-200 plastic scintillator (Eljen Technology)
- Readout: Hamamatsu PMT
- Manufacturer: Scionix Holland

Specification	Panel 0	Panel 1	Panel 2	Panel 3
Brand	Scionix Holland	Scionix Holland	Scionix Holland	Scionix Holland
Model	R280x10B280/2M-EI-P-X-NEG	R280x10B280/2M-EI-P-X-NEG	R280x10B280/2M-EI-P-X-NEG	R280x10B280/2M-EI-P-X-NEG
Scintillator	EJ-200	EJ-200	EJ-200	EJ-200
Serial No.	S2AA6047	S2AA6049	S2AA6050	S2AA6048
Type	Neg-Scintillator-Det	Neg-Scintillator-Det	Neg-Scintillator-Det	Neg-Scintillator-Det

| image_pmt0.png | image_pmt1.png | image_pmt2.png | image_pmt3.png |

Panel Voltage Settings

Each panel has a suggested maximum operating voltage provided by the manufacturer. The working voltages were set conservatively below the maximum limit to ensure stable long-term operation and to avoid damage to the PMTs.

Setting	Panel 0	Panel 1	Panel 2	Panel 3
Suggested Max Voltage	1560 V	1800 V	1340 V	1550 V
Set Max Voltage Limit	1460 V	1700 V	1240 V	1450 V
Working Voltage (V_set)	1400 V	1600 V	1150 V	1350 V

Data Acquisition (DAQ)

The signals from each panel are processed through a discriminator module, which converts the analog PMT pulses into digital logic signals for coincidence analysis.

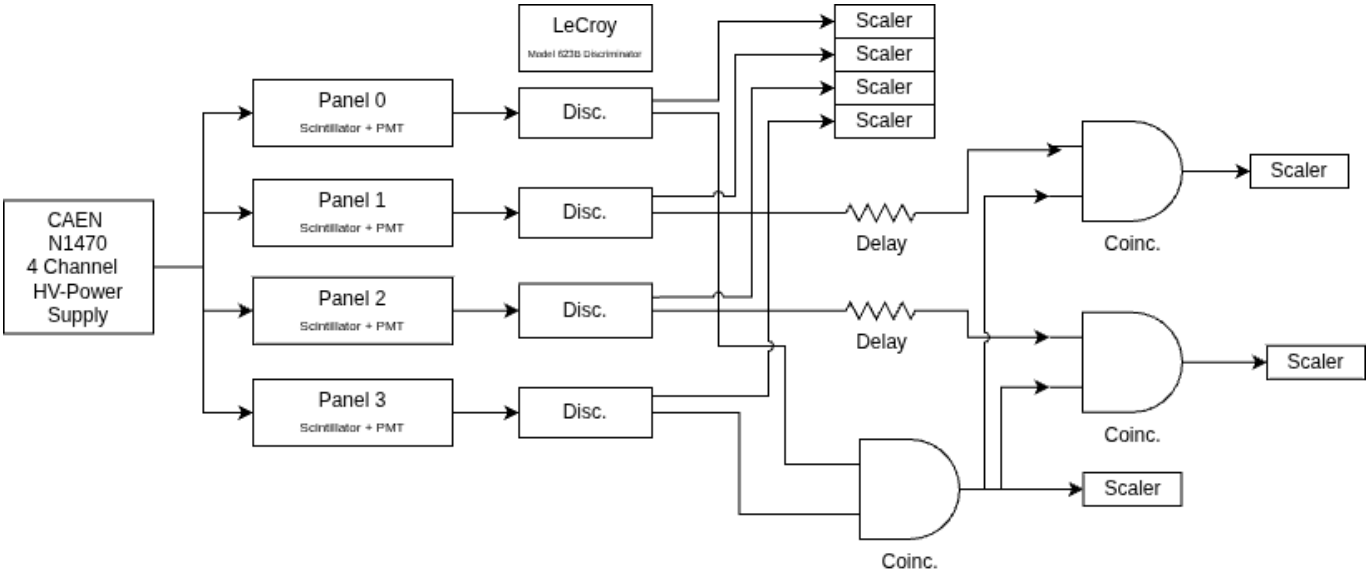
- Discriminator: LeCroy Model 623B



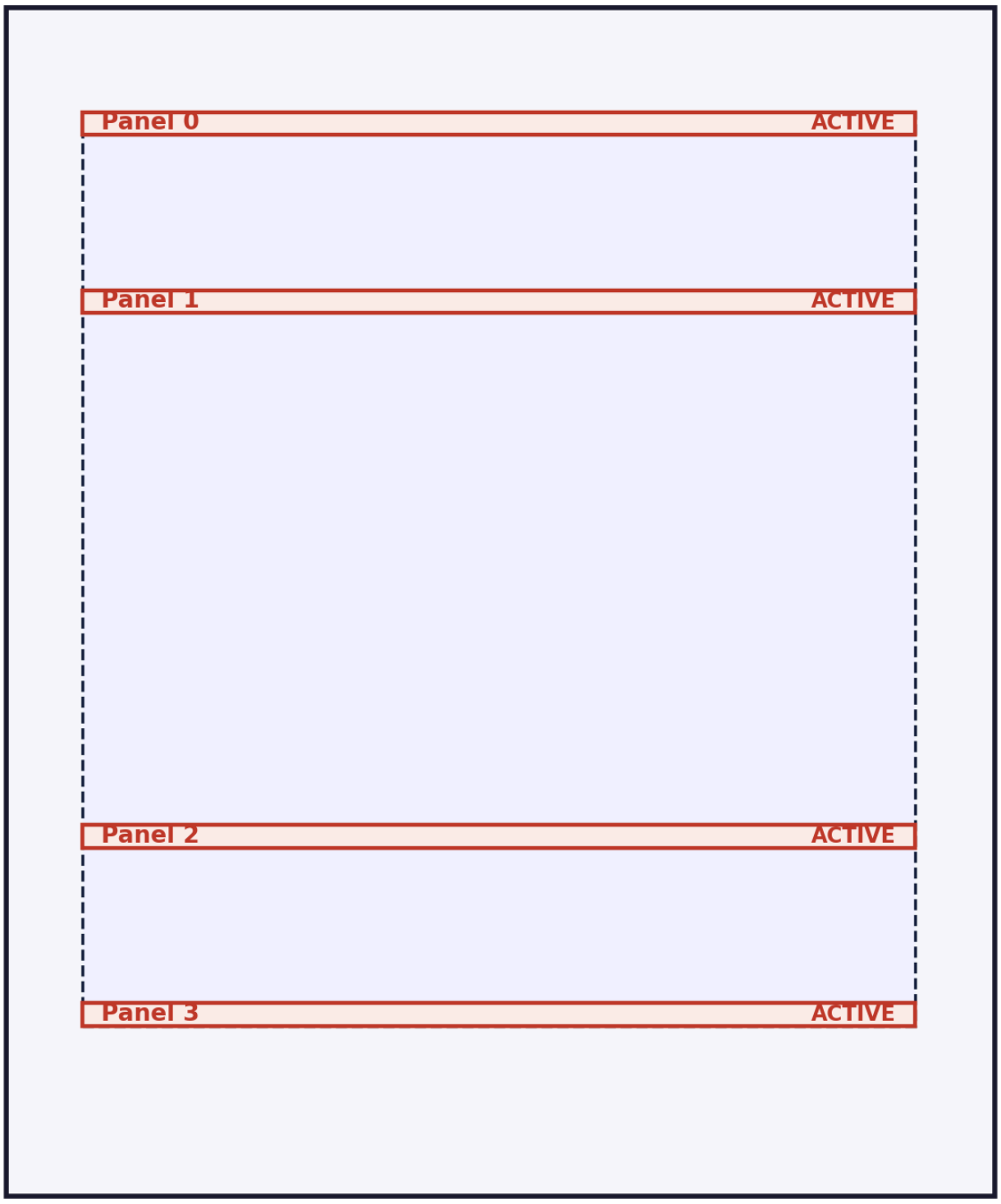
Measurement Plan

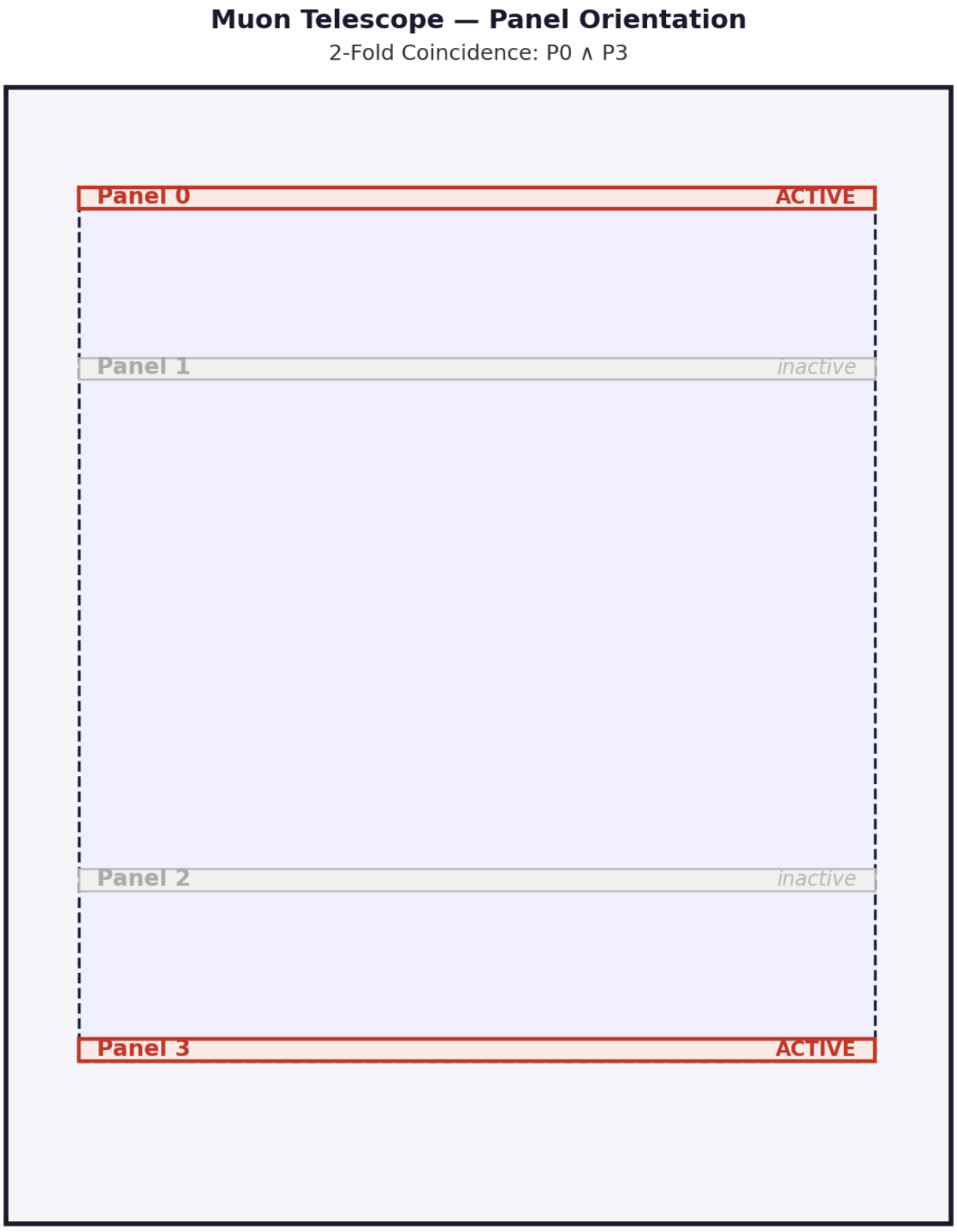
The DAQ chain operates as follows: PMT signals from each panel are fed into the discriminator, where a threshold is applied to reject noise. The discriminated outputs are then combined in a coincidence logic unit

to identify events where multiple panels fire within a defined time window.



Panel Layout


Muon Telescope — Panel Orientation
4-Fold Coincidence: P0 ∧ P1 ∧ P2 ∧ P3
Panel 0
ACTIVE
Panel 1
ACTIVE
Panel 2
ACTIVE
Panel 3
ACTIVE



|

First test results: 100 seconds for testing

Counts / Angle	θ = 0°	θ = −2.5°	θ = +2.5°	θ = 20°	θ = 45°
P0	1188	1212	1142	1179	—

Counts / Angle	θ = 0°	θ = −2.5°	θ = +2.5°	θ = 20°	θ = 45°
P1	1430	1427	1377	1402	—
P2	1066	1176	1066	1095	—
P3	1180	1280	1208	1344	—
P_all	28	20	22	43	—
P03	49	35	34	28	—

